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Space oven

Abstract

A space oven operates in microgravity environments by forcing convection towards the center through a unique heating element and airflow design. The space oven includes a tubular chamber, a heating rack, a heating system, a cooling system, a hatch, a user interface, a microcontroller, an enclosure, at least one first vent, at least one second vent and at least one temperature sensor. The tubular chamber is the cooking area. The heating rack holds consumables in place. The heating system heats up consumables. The cooling system prevents any overheating. The hatch closes off and allows access to the inside of the tubular chamber. The user interface allows a user to input commands. The microcontroller manages the electronic components. The enclosure protects the tubular chamber. The at least one first vent and the at least one second vent reduce pressure buildup. The at least one temperature sensor monitors the internal temperature.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
1334876	12/1919	Seymour	68/19	A47J 37/047
2288144	12/1941	Sickinger	219/403	F24C 15/168
2851029	12/1957	Fry et al.	N/A	N/A
3358120	12/1966	Schibley	219/403	F24C 15/166
4503760	12/1984	Pryputsch et al.	N/A	N/A
4516012	12/1984	Smith et al.	N/A	N/A
4928663	12/1989	Nevin et al.	N/A	N/A
4963091	12/1989	Hoetzl et al.	N/A	N/A
5075121	12/1990	Desage	426/233	F22B 1/30
6380524	12/2001	Keller	N/A	N/A
6513760	12/2002	Mueller et al.	N/A	N/A
6943322	12/2004	Whipple, Jr.	N/A	N/A
10512127	12/2018	Hayashi et al.	N/A	N/A
11950727	12/2023	Partiyeli	N/A	H05B 3/06
2007/0131670	12/2006	Nam et al.	N/A	N/A
2008/0047542	12/2007	Hughes	N/A	N/A
2008/0087657	12/2007	Lee et al.	N/A	N/A
2009/0025704	12/2008	Padula	N/A	N/A
2011/0266271	12/2010	Boyer et al.	N/A	N/A
2015/0330642	12/2014	Vasan	N/A	N/A
2016/0332785	12/2015	Sexton	N/A	B65D 51/1683
2016/0348918	12/2015	Bhogal et al.	N/A	N/A
2017/0265673	12/2016	Gustavsson et al.	N/A	N/A
2020/0037816	12/2019	Hackley	N/A	A47J 37/0664

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
104957963	12/2014	CN	N/A
19509569	12/1995	DE	N/A
19523520	12/1996	DE	N/A
102007043258	12/2008	DE	N/A
202011003427	12/2010	DE	A47J 37/048
0573599	12/1994	EP	N/A
20080076310	12/2007	KR	N/A
20080005014	12/2007	KR	N/A
20120016919	12/2011	KR	N/A
WO-9508288	12/1994	WO	A47J 37/047

OTHER PUBLICATIONS

KR20080005014U (Kim, Yungon) Oct. 29, 2008 [retrieved on Nov. 23, 2023]. Retrieved from Espacenet Database, translation by EPO and Google. (Year: 2008) (Year: 2008). cited by examiner
KR20080076310A (Kim, Kwang) Aug. 20, 2008 [retrieved on Dec. 3, 2023]. Retrieved from Espacenet Database, translation by EPO and Google. (Year: 2008) (Year: 2008). cited by examiner
DE-202011003427-U1 (Rennies Felix) May 5, 2011 [retrieved on Feb. 7, 2024]. Retrieved from Espacenet Database, translation by EPO and Google. (Year: 2011). cited by examiner
CN-104957963-A (Wang, Zhenyu) Oct. 7, 2015 [retrieved on Feb. 12, 2025]. Retrieved from Espacenet Database, translation by EPO and Google. (Year: 2005). cited by examiner

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This patent application is a continuation of and claims the benefit of priority to U.S. application Ser. No. 17/023,353, filed on Sep. 16, 2020, which claims a priority to U.S. Provisional Patent application Ser. No. 62/901,133 filed on Sep. 16, 2019, all of which are incorporated herein in their entireties by reference.

FIELD OF THE INVENTION
(1) The present invention generally relates to cooking appliances and microgravity utility devices. More specifically, the present invention provides a space oven intended to test how food items cook in microgravity environments.
BACKGROUND OF THE INVENTION
(2) An objective of the present invention is to provide an oven which is used to evaluate cooking in a zero-gravity environment. The lack of natural convection exceeding touch-safe temperature, and the limited food position control in zero gravity relative to oven-heating elements have restricted the use of conventional oven technology in space. The space oven of the present invention is provided as a stand-alone unit with all components pre-installed within an Express Rack Locker Insert without a door. A Velcro-on front cover panel is provided during cooking operations and cooldown to direct air flow across the oven and into the cooling rack area to cool sample trays.

Further, the front cover panel encloses sample trays in the cooling rack from exceeding touch temperatures when the space oven is unattended. The space oven is also externally powered and monitored. A digital display screen for internal oven temperature monitoring with menu controls is provided on an airflow plenum.
(3) The space oven provides restraint guiderails designed to provide controlled movement of the sample tray in and out of the oven and preclude contact with the heating elements or standoffs. Separate manual Master Power and Heater Power switches are provided to allow tests, diagnostics, or updates to the control system without powering the heating element. The heating element is sized so a worst-case interior oven temperature of 363 degrees Fahrenheit (° F.) is reached after two hours due to Aerogel insulation properties and continuous 28 Volts direct current (DC) power to the heating element provided from the EXPRESS Rack power supply. A door switch cuts power to the heating element and provides a signal to the controller if the oven door is not closed properly and displays a "DOOR OPEN" warning message on the liquid crystal display (LCD) screen. In addition, the fan controller cuts power to the heating element in event of fan failure/degradation. The present invention includes two independent resistance temperature detectors (RTDs) to measure the oven temperature and provide feedback to the controller to maintain normal operating temperature. Further, an overheat switch that cuts power to the heating element is provided which signals to the controller when the oven reaches a temperature of 392° F. (200 degrees Celsius), triggers a light-emitting diode (LED) light, and displays an "OVERHEAT" warning message on the LCD screen.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is an exploded perspective view of the present invention with the international-standard payload rack.
(2) FIG. **2** is a top perspective view of the present invention with the top of the enclosure being exposed.
(3) FIG. **3** is a top view of the present invention with the top of the enclosure being exposed and wherein the dashed arrows represent the first airflow path and the second airflow path.
(4) FIG. **4** is an exploded perspective view displaying the hatch detached from the tubular chamber.
(5) FIG. **5** is a front view of the tubular chamber with the hatch attached.
(6) FIG. **6** is a cross-section view taken along line **6-6** from FIG. **5** wherein the dashed line represents the at least one heating wire.
(7) FIG. **7** is a left-side view of the tubular chamber with the hatch attached.
(8) FIG. **8** is a cross-section view taken along line **8-8** from FIG. **7**.
(9) FIG. **9**A is a top perspective view of the heating tray.
(10) FIG. **9**B is a bottom perspective view of the heating tray.
(11) FIG. **10** is a schematic diagram illustrating the electronic connections of the present invention.
(12) FIG. **11** is a schematic diagram illustrating the electrical connections of the present invention.
DETAIL DESCRIPTIONS OF THE INVENTION
(13) All illustrations of the drawings are for the purpose of describing selected versions of the present invention and are not intended to limit the scope of the present invention.
(14) In reference to FIGS. **1** through **11**, the present invention is a space oven intended to test how food items cook in microgravity environments. In further detail, the present invention provides an oven that can operate in microgravity environments by forcing convection towards the center of a cooking chamber through a unique heating element arrangement and airflow design. A preferred embodiment of the present invention comprises a tubular chamber **1**, a heating rack **6**, a heating system **9**, a cooling system **13**, a hatch **17**, a user interface **18**, a microcontroller **19**, an enclosure

20, at least one first vent **24**, at least one second vent **25**, and at least one temperature sensor **26**. The tubular chamber **1** is the cooking area of the present invention. The heating rack **6** is used to hold consumables in place within the tubular chamber **1**. The heating system **9** provides heat in order to cook consumables placed within the tubular chamber **1**. The cooling system **13** prevents the present invention from overheating. The hatch **17** provides a means to close off and access the inside of the tubular chamber **1**. The user interface **18** allows a user to input commands in order to heat up consumables. The microcontroller **19** controls and manages the electronic components of the present invention. The enclosure **20** protects the tubular chamber **1** and is designed to establish a unique airflow. The at least one first vent **24** and the at least one second vent **25** prevent the tubular chamber **1** from being a closed system and reduce pressure buildup within the tubular chamber **1**. The at least one temperature sensor **26** monitors the temperature within the tubular chamber **1**.

(15) The general configuration of the aforementioned components allows the present invention to operate in microgravity environments by forcing convection towards the center of the tubular chamber **1** through a unique heating element arrangement and airflow design. With reference to FIG. **4**, the tubular chamber **1** comprises an open chamber end **2** and a closed chamber end **3**. The tubular chamber **1** is preferably a double walled cylindrical chamber machined form aluminum and stainless steel. With reference to FIGS. **6** and **8**, the heating rack **6** is mounted within the tubular chamber **1** and is centrally positioned along the tubular chamber **1**. In further detail, the heating rack **6** is preferably mounted through a set of angled mounting hole. Further, this arrangement prevents the heating rack **6** from rotating within the tubular chamber **1** and allows the heating rack **6** to receive optimal heat from the heating system **9**. The heating system **9** is also mounted within tubular chamber **1** and is distributed around the heating rack **6**. In further detail, the heating system **9** is preferably mounted through a set of fasteners which are positioned around the heating rack **6**. This arrangement distributes heat towards the center of the tubular chamber **1**, and therefore, heating is applied to all sides of the heating rack **6**. Moreover, standard conditions, air near the inner lateral surface of the tubular chamber **1** would transfer heat to the outer lateral surface of the tubular chamber **1**, and that heat will be conducted away to the exterior of the present invention and lost to the environment. However, the arrangement of the heating rack **6**, the heating system **9**, and the tubular chamber **1** must accommodate for the lack of convection in microgravity. In microgravity conditions, the low thermal conductivity of air will limit the heat from flowing from the pocket of hot air at the center of the tubular chamber **1** to the lateral surface of the tubular chamber **1**. Thus, this arrangement allows consumables, placed within the tubular chamber **1**, to be exposed to cooking temperatures, while the lateral surface of the tubular chamber **1** remains relatively cool and safe to touch.

(16) In addition and with reference to FIGS. **2** through **4**, the hatch **17** is operatively mounted to the open chamber end **2**. The hatch **17** is used to selectively access the open chamber end **2**. In further detail, the hatch **17** includes a matching hollow design with a center opening to receive a temperature gauge. The hatch **17** is preferably mounted to the open chamber end **2** with a plurality of O-rings provided in between to seal the connection. Further, the hatch **17** may include a handle to facilitate the safe opening and closing of the hatch **17**. With reference to FIG. **6**, the at least one first vent **24** is integrated into the hatch **17**. In further detail, the at least one first vent **24** is preferably an installation vent that traverses through the hatch **17** in order to prevent pressure buildup during operation of the present invention. Similarly, the at least one second vent **25** is integrated into the closed chamber end **3**. The at least one second vent **25** is preferably an installation vent, which is more specifically a 40-micron porous stainless-steel vent. The installation vent of the at least one first vent **24** and the installation vent of the at least one second vent **25** work together to provide a continuous vent design. Therefore, no cracking pressure is required to active the present invention. Further, the 40-micron porous stainless-steel vent of the at least one second vent **25** releases pressure from the tubular chamber **1** when gases expanded during

operation of the present invention.

(17) Moreover and with reference to FIGS. **3** and **10**, the tubular chamber **1**, the microcontroller **19**, and the user interface **18** are mounted within the enclosure **20**. Thus, the tubular chamber **1**, the microcontroller **19**, and the user interface **18** are protected by the enclosure **20** and held in place within the enclosure **20**. In further detail, the tubular chamber **1** is preferably mounted within the enclosure **20** through at least one fastening bracket. The cooling system **13** is integrated into the enclosure **20**. In further detail, the cooling system **13** traverses through the backend of the enclosure **20** to direct cool air into the enclosure **20** while simultaneously pulling hot air out of the enclosure **20**. The at least one temperature sensor **26** is mounted into the closed chamber end **3**. In further detail, the at least one temperature sensor **26** is preferably mounted through a hole in the closed chamber end **3** by a set of fasteners. Moreover, this arrangement is sealed with a high-temperature room temperature vulcanizing (RTV) sealant.

(18) Furthermore and with reference to FIG. **10**, the microcontroller **19** is electronically connected to the heating system **9**, the cooling system **13**, the user interface **18**, and the at least one temperature sensor **26**. Thus, the microcontroller **19** can manage and control the heating system **9**, the cooling system **13**, the user interface **18**, and the at least one temperature sensor **26**. The microcontroller **19** is preferably an AT Mega 32U4 microcontroller **19**. The user interface **18** can be any type of user interface **18** such as, but not limited to, a touch-screen display screen or a liquid crystal display (LCD) screen with a plurality of control buttons. Inputs from the user interface **18** are monitored by the microcontroller **19**, and a context dependent menu is displayed on the user interface **18**. Nominal operation of the present invention begins with turning on the “Master Power” and “Heater Power” switches of the user interface **18**. The LCD screen of the user interface **18** shows a context dependent menu that allows the operator to use the three pushbuttons of the user interface **18** to cycle through and select options. A user can select a “Preheat” temperature, or a “Cook” temperature and cooking time, or enter an options menu to change oven settings. Oven temperature can be set in 5-degree increments, and time can be set in one-minute increments. Once an oven temperature is set by the user, the microcontroller **19** operates as a two-state controller to cycle the heating system **9** on and off to maintain the internal temperature of the tubular chamber **1**. A+3° hysteresis is built into the control program to prevent the microcontroller **19** from switching at high frequency when the present invention reaches its set temperature. The microcontroller **19** monitors the internal temperature of the tubular chamber **1** via the at least one temperature sensor **26** which preferably includes a set of platinum resistive temperature devices (RTDs) and controls a solid-state relay (SSR) to apply power to the heating system **9**. The at least one temperature sensor **26** may also include an automated shutoff thermostat that automatically disables the heating system **9** if a high-temperature threshold is reached. The microcontroller **19** monitors and manages the air temperature in the enclosure **20** around the user interface **18** and the tubular chamber **1** by the cooling system **13**.

(19) With reference to FIG. **6**, the present invention may further comprise a quantity of insulation material **27** in order to prevent the release of heat from the inside of the tubular chamber **1**. As mentioned previously, the tubular chamber **1** is preferably a double-walled cylindrical chamber and, thus, comprises an inner lateral wall **4** and an outer lateral wall **5**. The inner lateral wall **4** is concentrically encircled by the outer lateral wall **5**. The quantity of insulation material **27** is preferably compressed granulated Aerogel insulation and is positioned between the inner lateral wall **4** and the outer lateral wall **5**. Thus, this arrangement prevents the release of heat from the inside of the present invention. Similarly, the hatch **17** may also be doubled-walled like the tubular chamber **1** and filled with the same insulation.

(20) In order for the heating system **9** to effectively distribute heat towards the center of the tubular chamber **1** and with reference to FIG. **6**, the heating system **9** comprises a plurality of first standoffs **10**, a plurality of second standoffs **11**, and at least one heating wire **12**. The plurality of first standoffs **10** is mounted adjacent to the open chamber end **2**. In further detail, the plurality of

first standoffs **10** is preferably a set of ceramic standoffs that are recessed into aluminum bosses and encapsulated with a high-temperature RTV sealant to prevent debris from being released into a space station environment in case of accidental shattering of the ceramic material. Fasteners such as stainless-steel screws are mounted in each of the plurality of first standoffs **10**. Similarly, the plurality of second standoffs **11** is mounted adjacent to the closed chamber end **3**. In further detail, the plurality of second standoffs **11** is preferably a set of ceramic standoffs that are recessed into aluminum housings and encapsulated with a high-temperature RTV sealant to prevent debris from being released into a space station environment in case of accidental shattering of the ceramic material. Fasteners such as stainless-steel screws are mounted in each of the plurality of second standoffs **11**. The fastener of the plurality of first standoffs **10** and the fasteners of the plurality of second standoffs **11** provide electrically insulated mounting points for the at least one heating wire **12**. The at least one heating wire **12** is preferably a nichrome **80** wire. The plurality of first standoffs **10** and the plurality of second standoffs **11** are distributed around the heating rack **6**, and the at least one heating wire **12** is strung in between the plurality of first standoffs **10** and the plurality of second standoffs **11**. Thus, this arrangement effectively distributes heat towards the center of the tubular chamber **1**.

(21) In order for the cooling system **13** to effectively prevent the present invention from overheating and with reference to FIG. **2**, the cooling system **13** comprises an inlet fan assembly **14**, an outlet fan assembly **15**, and a plurality of airflow-guiding panels **16**. The inlet fan assembly **14** and the outlet fan assembly **15** are positioned adjacent to the closed chamber end **3**. In further detail, the inlet fan assembly **14** and the outlet fan assembly **15** are preferably mounted into the backend of the enclosure **20**. The inlet fan assembly **14** is positioned offset from the outlet fan assembly **15**. The inlet fan assembly **14** is in fluid communication with the outlet fan assembly **15** through the plurality of airflow-guiding panels **16**, the at least one first vent **24**, and the at least one second vent **25**. In further detail, the inlet fan assembly **14** directs cool air into the enclosure **20** through the air-flow guiding panels **16**, and the outlet fan assembly **15** pulls hot air out of the enclosure **20** while the at least one first vent **24** and the at least one second vent **25** slowly reduce pressure from the inside of the tubular chamber **1** by letting out gas. With reference to FIG. **3**, a first airflow path is delineated from the inlet fan assembly **14**, in between the open chamber end **2** and the closed chamber end **3** (i.e. about the lateral surface of the tubular chamber **1**) and to the outlet fan assembly **15**. The first airflow path provides a forced convection over the tubular chamber **1**. A second airflow path is delineated from the inlet fan assembly **14**, across the hatch **17**, and to the outlet fan assembly **15**. The second airflow path cools the user interface **18** and the microcontroller **19**. In the preferred embodiment, the user interface **18** is mounted onto one of the plurality of airflow-guiding panels **16**, and the microcontroller **19** is mounted within the same airflow-guiding panel. Thus, the tubular chamber **1** and the electronic components of the present invention can be cooled by the cooling system **13**. The inlet fan assembly **14** and the outlet fan assembly **15** may each include a fan controller that monitors the air temperature inside the enclosure **20** around the electronic components of the present invention and the tubular chamber **1**. The fan controller reports fan failures or off-nominal temperature conditions to the microcontroller **19**.

(22) With reference to FIGS. **2** and **3**, the present invention may further comprise at least one cooling rack **28** in order to allow consumables to cool after being heated by the present invention. The at least one cooling rack **28** is mounted within the enclosure **20** and is positioned adjacent to the open chamber end **2**. In further detail, the at least one cooling rack **28** is preferably mounted in between two of the plurality of airflow-guiding panels **16**. The first airflow path and the second airflow path are intersected by the at least one cooling rack **28**. In further detail, cool air is drawn into an inlet airflow-guiding panel from the plurality of airflow-guiding panels **16** by the inlet fan assembly **14** and directed over the electronic components of the present invention. The cool air is then directed by vents to flow over the tubular chamber **1**, the hatch **17**, and the at least one cooling

rack **28**.

(23) With reference to FIG. **8**, the present invention may further comprise a heating tray **29** in order to safely heat consumables within the tubular chamber **1**. The heating rack **6** comprises a first track **7** and a second track **8**. The first track **7** and the second track **8** are each preferably a set of stainless-steel tubing. The first track **7** and the second track **8** traverse from the open chamber end **2** to the closed chamber end **3**. Further, the first track **7** and the second track **8** are positioned opposite to each other within the tubular chamber **1**. This arrangement prevents consumables, placed on the heating rack **6**, from rotating within the tubular chamber **1**. The heating tray **29** is slidably engaged in between the first track **7** and the second track **8**. Thus, the heating tray **29** is held in place by the heating rack **6**.

(24) In order for the heating rack **6** to safely seal consumables in preparation to be heated by the present invention and with reference to FIGS. **9**A and **9**B, the heating tray **29** may further comprise a first sealing assembly **30** and a second sealing assembly **31**. The first sealing assembly **30** and the second sealing assembly **31** each comprise a flexible sheet **32**, a rigid frame **33**, and a pressure-release valve **34**. The flexible sheet **32** is preferably a silicone sheet. The rigid frame **33** is preferably a rectangular, aluminum frame. The pressure-release valve **34** is preferably a 40-micron vent. The pressure-release valve **34** is integrated into the flexible sheet **32**. This arrangement prevents pressure build within the heating tray **29**. The rigid frame **33** is perimetrically connected around the flexible sheet **32**. This arrangement allows the rigid frame **33** to border the flexible sheet **32**. Further, the rigid frame **33** of the first sealing assembly **30** is hermetically connected to the rigid frame **33** of the second sealing assembly **31**. This arrangement ensures an air-tight seal for the heating tray **29** to package cookable consumables. Moreover, the rigid frame **33** allows the heating tray **29** to be slidably engaged in between the first track **7** and the second track **8**.

(25) In order for the enclosure **20** to protect the tubular chamber **1** and the electronic components of the present invention while still allowing access to the tubular chamber **1** and the electronic components of the present invention and with reference to FIG. **1**, the enclosure **20** comprises a receptacle **21** and a cover **23**. The receptacle **21** comprises an opening **22**. With reference to FIG. **2**, the open chamber end **2**, the user interface **18**, and the microcontroller **19** are positioned adjacent to the opening **22**. This arrangement allows a user to easily access the inside of the tubular chamber **1** and the user interface **18**. Further, the cooling system **13** is positioned opposite to the opening **22** about the enclosure **20**. This allows the cooling system **13** to easily direct cool air into and pull hot air from the enclosure **20**. The receptacle **21** is perimetrically attached to the cover **23**, and the cover **23** is positioned across the opening **22**. The cover **23** can be perimetrically attached to the cover **23** by any fastening mechanism but is preferably attached to the cover **23** through a hook-and-loop fastener. Thus, the tubular chamber **1** and the user interface **18** can be accessed by detaching the cover **23** from the receptacle **21**.

(26) In order to provide a safety measure to the operation of the present invention and with reference to FIG. **10**, the present invention may further comprise a door switch **35**. The door switch **35** is operatively integrated in between the hatch **17** and the open chamber end **2**. The door switch **35** is used to detect a closed configuration for the hatch **17** and an open configuration for the hatch **17**. The microcontroller **19** is electronically connected to the door switch **35**. In further detail, the door switch **35** shuts off power to the heating system **9** when a user opens the hatch **17** to insert or remove consumables or in case the hatch **17** is not closed and secured properly.

(27) In order for the microcontroller **19** to be monitored and updated and with reference to FIG. **10**, the present invention may further comprise a communication computing device **36**. The communication computing device **36** is preferably a BeagleBone black single-board computer that monitors the payload and provides an interface for ground controllers to interact with the present invention. The communication computing device **36** monitors the fan controllers for fan failures or off nominal temperature conditions and records the plenum temperatures to a log file. The communication computing device **36** is mounted within the enclosure **20** and is electronically

connected to the microcontroller **19**. This arrangement allows the communication computing device **36** to interface with the microcontroller **19** and allows updates and new operational programs to be uploaded to the microcontroller **19**.
(28) In order for the present invention to communicate and receive electrical power from a space station and with reference to FIGS. **10** and **11**, the present invention may further comprise a data port **37**, a power port **38**, and an international-standard payload rack **39**. The data port **37** and the power port **38** are integrated into the enclosure **20**. In further detail, the data port **37** and the power port **38** are standard power and ethernet connections for devices to interface with a space station. The enclosure **20** is mounted into the international-standard payload rack **39** through a set of fasteners in order for the international-standard payload rack **39** to protect and hold the enclosure **20** in place. The international-standard payload rack **39** is electronically connected to the communication computing device **36** by the data port **37**. This allows data to be relayed between the present invention, through the international-standard payload rack **39**, and a space station. The international-standard payload rack **39** is electrically connected to the heating system **9**, the cooling system **13**, the user interface **18**, the microcontroller **19**, the at least one temperature sensor **26**, and the communication computing device **36** by the power port **38**. This allows 28 Volts direct current (DC) power to be supplied to the present invention via the international-standard payload rack **39**. The electrical power from the space station is conditioned by a power supply of the present invention to provide 28 Volts DC power to the heating system **9** and 5 Volts DC power to the cooling system **13**, the user interface **18**, the microcontroller **19**, the at least one temperature sensor **26**, and the communication computing device **36**.
(29) Although the invention has been explained in relation to its preferred embodiment, it is to be understood that many other possible modifications and variations can be made without departing from the spirit and scope of the invention as hereinafter claimed.

Claims

1. A space oven for operation in a micro-gravity environment, comprising: a chamber that includes a first end opposite a second end along a central axis; at least one heating rack mounted within the chamber along the central axis; at least one heating wire, each heating wire fastened to a first plurality of locations surrounding the central axis on the first end and a second plurality of locations surrounding the central axis on the second end, so that each heating wire is fastened to the chamber at locations that alternate between the first end and the second end, such that operating the at least one heating wire in a micro-gravity environment causes heat to be conveyed toward the central axis; and a heating tray that, in a closed position, defines a hermetically sealed interior, and that is removably mountable in the chamber via the at least one heating rack.
2. The space oven of claim 1, further comprising: at each of the first plurality of locations and each of the second plurality of locations, a respective standoff that offsets the at least one heating wire away from the first and second ends, respectively.
3. The space oven of claim 2, wherein, in each case, the respective standoff is recessed into a boss and encapsulated with a sealant.
4. The space oven of claim 2, further comprising: a respective fastener that, in each case, mounts the at least one heating wire to the respective standoff.
5. The space oven of claim 1, wherein the first plurality of locations and the second plurality of locations define electrically insulated mounting points for the at least one heating wire.
6. The space oven of claim 1, wherein: the first end includes an open end; and the space oven further comprises a hatch operatively mounted to the open end.
7. The space oven of claim 1, wherein the at least one heating wire is arranged so as to form a spacing of air that is between the heating wire and an interior surface of the chamber and that surrounds the central axis, such that a flow of conducted heat resulting from operation of the

heating wire is biased toward the central axis and so as to inhibit heating of the interior surface of the chamber.
8. A method of operating a space oven in a micro-gravity environment, comprising: with the space oven positioned in a micro-gravity environment, inserting at least one heating tray into a chamber of the space oven via a first end opposite a second end along a central axis, and removably mounting the at least one heating tray in at least one heating rack mounted within the chamber along the central axis; and operating at least one heating wire, each heating wire fastened to a first plurality of locations surrounding the central axis on the first end and a second plurality of locations surrounding the central axis on the second end, so that each heating wire is fastened to the chamber at locations that alternate between the first end and the second end, such that heat is conveyed toward the central axis.
9. The method of claim 8, wherein, at each of the first plurality of locations and the second plurality of locations, a respective standoff offsets the at least one heating wire away from the first and second ends, respectively.
10. The method of claim 9, wherein, in each case, the respective standoff is recessed into a boss and encapsulated with a sealant.
11. The method of claim 9, wherein, in each case, a respective fastener mounts the at least one heating wire to the respective standoff.
12. The method of claim 8, wherein the first plurality of locations and the second plurality of locations define electrically insulated mounting points for the at least one heating wire.
13. The space oven of claim 8, wherein: the first end includes an open end; and the space oven further comprises a hatch operatively mounted to the open end.
14. The space oven of claim 8, further comprising: inserting at least one item to be heated into an interior of the at least one heating tray in an open position; and transitioning the at least one heating tray into a closed position, such that the interior is hermetically sealed.
15. The method of claim 8, wherein the at least one heating wire is arranged so as to form a spacing of air that is between the heating wire and an interior surface of the chamber and that surrounds the central axis, such that a flow of conducted heat resulting from operation of the heating wire is biased toward the central axis and so as to inhibit heating of the interior surface of the chamber, such that the interior surface remains safe to human touch during operation of the at least one heating wire.
16. A method of producing a space oven, comprising: mounting at least one heating rack within a chamber of the space oven along a central axis of the chamber that extends between a first end and a second end opposite the first end; fastening at least one heating wire such that each heating wire is fastened to a first plurality of locations surrounding the central axis on the first end and a second plurality of locations surrounding the central axis on the second end, so that the each wire is fastened to the chamber at locations that alternate between the first end and the second end, such that operating the at least one heating wire in a micro-gravity environment causes heat to be conveyed toward the central axis; and removably mounting a heating tray in the chamber via the at least one heating rack, wherein the heating tray, in a closed position, defines a hermetically sealed interior.
17. The method of claim 16, wherein fastening the at least one heating wire to the first plurality of locations and the second plurality of locations includes: recessing a respective standoff into a boss at each of the first plurality of locations and the second plurality of locations; in each case, fastening the at least one heating wire to the respective standoff, such that the at least one heating wire is offset from the first and second ends, respectively, by the respective standoff; and encapsulating the respective standoff with a sealant.
18. The method of claim 17, wherein: in each case, the at least one heating wire is fastened to the respective standoff via a respective fastener; and the first plurality of locations and the second plurality of locations define electrically insulated mounting points for the at least one heating wire.

19. The space oven of claim 16, wherein the at least one heating wire is arranged so as to form a spacing of air that is between the heating wire and an interior surface of the chamber and that surrounds the central axis, such that a flow of conducted heat resulting from operation of the heating wire is biased toward the central axis and so as to inhibit heating of the interior surface of the chamber.